

SatoshiFlow

Algorithmic Trading Strategies for Bitcoin (BTC/USD)

Summer of Quant

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1 Strategy Logic & Hypothesis

Given Bitcoin's historical tendency to exhibit strong, prolonged directional trends, my primary hypothesis is that trend-following and momentum breakout strategies will significantly outperform mean-reversion strategies (I tried one and failed miserably). To validate this, three distinct trend-following strategies were developed. All strategies incorporate a dynamic volatility-based trailing stop-loss utilizing the Average True Range (ATR) to lock in profits and cut losses during volatile market swings.

1.1 Strategy 1: EMA Trend-Following (The Primary Strategy)

The core strategy relies on the crossover of a fast Exponential Moving Average (EMA) and a slow EMA to identify shifting trends.

- **Hypothesis/Logic:** A LONG position is initiated when the Fast EMA crosses above the Slow EMA, indicating bullish momentum. Conversely, a SHORT position is initiated when the Fast EMA crosses below the Slow EMA.
- **Filtering:** To prevent buying at the top of a trend, the Relative Strength Index (RSI) is used as a filter. LONG trades are only executed if the 14-day RSI is below 70, and SHORT trades are executed if the RSI is above 30.
- **Parameters:** Through grid-search optimization, the ideal parameters were found to be a 25-day Fast EMA, a 75-day Slow EMA, and a 2.5x ATR trailing stop.

1.2 Strategy 2: Donchian Channel Breakout

Often referred to as "Turtle Trading," this strategy capitalizes on the assumption that breaking into a new high or low signifies the start of a major trend.

- **Hypothesis/Logic:** A LONG position is opened when the daily closing price exceeds the highest high of the past 20 days. A SHORT position is opened when the price drops below the lowest low of the past 20 days.
- **Exit Logic:** Positions are closed when the price breaks the 10-day low (for longs) or the 10-day high (for shorts).

1.3 Strategy 3: MACD Momentum

This strategy uses the Moving Average Convergence Divergence (MACD) indicator to capture short-term momentum shifts.

- **Entry Logic:** A LONG position is initiated when the MACD line (the difference between the 12-day and 26-day EMA) crosses above its 9-day Signal Line. A SHORT position is initiated when the MACD line crosses below its Signal Line.
- **Exit Logic:** Positions are closed dynamically using a 2.5x ATR trailing stop to lock in profits while limiting downside exposure.
- **Note on RSI Filtering:** While an RSI filter improved the other strategies, it was detrimental here. Because the MACD is a lagging indicator, by the time a crossover signaled a trade, the RSI was often already indicating overbought/oversold conditions, causing the filter to block almost all valid breakout trades.

2 Key Performance Metrics

All strategies were backtested using the provided `backtester.py` framework, factoring in a starting capital of \$1,000 and 0.15% trading fees. The strategies were verified to be completely free of lookahead bias.

Metric	EMA Trend (25/75)	Donchian Breakout	MACD Momentum
Sharpe Ratio	1.147	0.922	0.760
Net Profit	\$5,597.58	\$5,334.30	\$3,237.76
Win Rate	50.00%	44.18%	36.58%
Total Trades	48	43	123
Max Drawdown	24.71%	45.88%	52.53%

Table 1: Performance metrics of the three tested strategies over the 2018-2022 dataset.

3 Comparison and Final Rankings

The backtesting results strongly validate our initial hypothesis: Bitcoin is highly receptive to long-term trend-following strategies.

1. **1st Place - Optimized EMA Trend-Following:** This strategy significantly outperformed the others. By utilizing an RSI filter to avoid late entries and an ATR stop to manage risk, it yielded the highest Sharpe Ratio (1.147), the highest net profit (\$5,597.58), and the lowest Maximum Drawdown (24.71%).
2. **2nd Place - Donchian Channel Breakout (with RSI Filter):** This strategy performed exceptionally well as a breakout system, especially after adding an RSI filter to avoid buying into overextended trends. It yielded a massive net profit (\$5,334.30) and an excellent Sharpe Ratio of 0.922, though its drawdowns (45.88%) were higher than the EMA strategy due to delayed exit signals.

3. **3rd Place - MACD Momentum:** This strategy generated too many false signals (whipsaws) in sideways markets. This led to over-trading (123 total trades), a low win rate (36.58%), and high commission costs, placing it in last with a Sharpe Ratio of 0.760.

Ultimately, pairing moving average crossovers with an RSI filter and ATR trailing stops proved to be the most effective, risk-adjusted method for algorithmic trading in this cryptocurrency market.